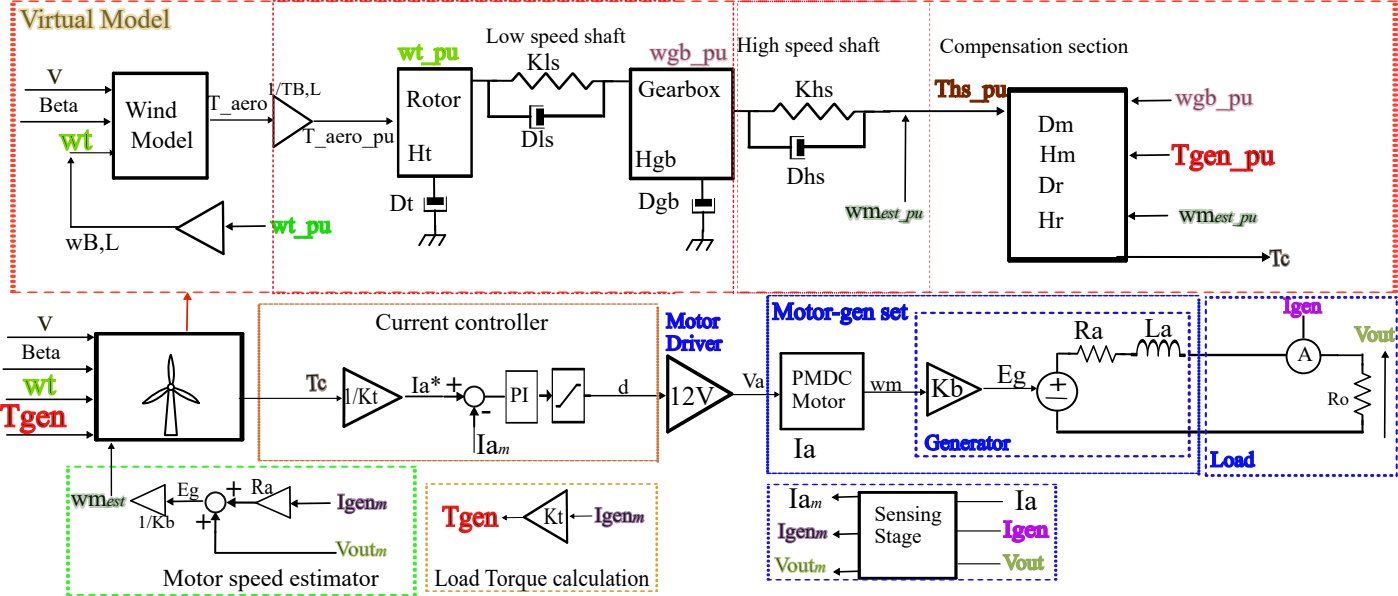

Virtual Model
V
Beta
wt
Wind Model
T_aero
1/TB,L
T_aero_pu
wB,L
wt_pu
wt_pu
Rotor
Ht
Dt
Low speed shaft
Kls
Dls
wgb_pu
Gearbox
Hgb
Dgb
High speed shaft
Khs
Dhs
wmest_pu
Compensation section
Ths_pu
Dm
Hm
Dr
Hr
wgb_pu
Tgen_pu
wmest_pu
Tc
V
Beta
wt
Tgen
Current controller
Tc
1/Kt
Ia*
PI
d
Iam
Motor Driver
12V
Va
Motor-gen set
PMDC Motor
Ia
wm
Kb
Eg
Ra
La
Generator
Igen
Vout
Ro
Load
wmest
Eg
Ra
1/Kb
Igenm
Voutm
Motor speed estimator
Tgen
Kt
Igenm
Load Torque calculation
Iam
Igenm
Voutm
Sensing Stage
Ia
Igen
Vout